EXP-9 TEXT GENERATION USING LSTM NETWORKS

AIM:

To implement a Long Short-Term Memory (LSTM) network for generating text corpus

SOFTWARE REQUIREMENT:

* Python 3.x
* TensorFlow/Keras
* Numpy
* Shakespeare Text corpus

ALGORITHM

* Import TensorFlow and Numpy
* Load the Shakespeare text corpus
* Convert characters into numerical indices
* Create input and target sequences
* Construct an LSTM model
* Compile the model on the ~~text sequences~~ Adam optimizer
* Train the model on the text sequences
* Provide a seed sentence
* Predict subsequent characters
* Display the generated text.

PROGRAM:

```
import tensorflow as tf
import numpy as np

text = tf.keras.utils.get_file(
        "shakespeare.txt",
        "https://raw.githubusercontent.com/karpathy/"
        "char-rnn/master/data/tinyshakespeare/input.txt")
```

EXPECTED OUTPUT

Generated Text:

to be a not be ...

```
text = open(text, encoding="utf-8").read().lower()
chars = sorted(set(text))
c2i = {c: i for i, c in enumerate(chars)}
i2c = {i: c for i, c in enumerate(chars)}

seq_len = 40
x = []
y = []
for i in range(len(text) - seq_len):
    x.append([c2i[c] for c in text[i:i+seq_len]])
    y.append(c2i[text[i+seq_len]])

x = np.array(x)
y = np.array(y)

def generate(seed, length=200):
    seq = [c2i[c] for c in seed.lower()]
    for _ in range(length):
        inp = np.array(seq[-seq_len:]).reshape(1, -1)
        pred = model.predict(inp, verbose=0)[0]
        next_char = np.random.choice(len(pred), p=pred)
        seq.append(next_char)
    return "".join(i2c[i] for i in seq)
print("Generated Text:")
print(generate("to be or not to be", 200))
```

RESULT:

An LSTM-based text generation model was implemented to learn character-level patterns from the Shakespeare corpus and generate new text sequences.